

Users, Groups, and Access Control

Users, Groups, and Access Control

Learning Objectives

- Understand Linux users and groups
 - Manage users and groups
 - Understand file and directory ownership
 - Configure file and directory permissions
 - Use numeric (octal) and symbolic permission methods
 - Apply special permissions (SUID, SGID, Sticky Bit)
 - Implement access control using ACLs
 - Troubleshoot permission-related issues
-

1. Introduction to Linux Security

Linux is a multi-user operating system where multiple users can access the same system simultaneously.

To protect files and resources, Linux uses:

1. Users
2. Groups
3. Permissions
4. Ownership
5. Access Control Lists (ACLs)

These mechanisms ensure that only authorized users can access specific files and directories.

2. Understanding Users

Every person using a Linux system has a user account.

Types of Users

1. Root User

- Super administrator
- Has complete control over the system

- User ID (UID): 0

Example:

```
whoami
```

Output:

```
root
```

2. Regular User

Used for daily operations.

Example:

```
meet  
john  
student1
```

Check current user:

```
whoami
```

3. System Users

Created automatically for services.

Examples:

```
www-data  
mysql  
sshd  
nobody
```

View all users:

```
cat /etc/passwd
```

3. Understanding Groups

Groups help manage permissions for multiple users.

Example:

```
Developers  
Students
```

HR
Finance

Instead of assigning permissions individually, permissions can be assigned to a group.

View groups:

```
cat /etc/group
```

Current user's groups:

```
groups
```

4. User Management Commands

Create User

```
sudo useradd -m student1
```

Set password:

```
sudo passwd student1
```

Create User with Home Directory

```
sudo adduser student2
```

Check User Information

```
id student1
```

Example Output:

```
uid=1001(student1)  
gid=1001(student1)  
groups=1001(student1)
```

Switch User

```
su - student1
```

Delete User

```
sudo userdel -r student1
```

5. Group Management Commands

Create Group

```
sudo groupadd cyber
```

Add User to Group

```
sudo usermod -aG cyber student1
```

Verify Group Membership

```
groups student1
```

Remove User from Group

```
sudo gpasswd -d student1 cyber
```

Delete Group

```
sudo groupdel cyber
```

6. File Ownership

Every file in Linux has:

- Owner
- Group

Check ownership:

```
ls -l
```

Example:

```
-rw-r--r-- 1 meet cyber 500 Jun 11 file.txt
```

Meaning:

Owner = meet

Group = cyber

7. Change Ownership

Change Owner

```
sudo chown student1 file.txt
```

Change Owner and Group

```
sudo chown student1:cyber file.txt
```

Change Group

```
sudo chgrp cyber file.txt
```

8. Understanding Permissions

View permissions:

```
ls -l
```

Example:

```
-rwxr-xr--
```

Breakdown:

Owner Group Others

rwx r-x r--

Permission Meaning

Permission	Value	Meaning
r	4	Read
w	2	Write
x	1	Execute

9. Numeric Permission Method

Common Permissions

Permission	Number
rwX	7
rw-	6
r-X	5
r--	4
---	0

Examples

777

```
chmod 777 file.txt
```

```
rwX rwX rwX
```

Everyone has full access.

755

```
chmod 755 script.sh
```

```
rwX r-X r-X
```

Owner full access.

Others read and execute.

644

```
chmod 644 file.txt
```

```
rw- r-- r--
```

Most common file permission.

600

```
chmod 600 secret.txt
```

```
rw-----
```

Only owner can access.

10. Symbolic Permission Method

Add Permission

```
chmod u+x file.sh
```

Add execute permission for owner.

Remove Permission

```
chmod g-w file.txt
```

Remove write permission from group.

Grant Permission

```
chmod o+r file.txt
```

Add read permission for others.

Multiple Changes

```
chmod u+rw,g+rx,o-r file.txt
```

11. Directory Permissions

Create directory:

```
mkdir project
```

Check permissions:

```
ls -ld project
```

Directory Permissions Meaning

Permission	Meaning
r	List contents
w	Create/Delete files
x	Enter directory

Example:

```
chmod 755 project
```

12. Real-World Lab Practical

Create Users

```
sudo useradd -m alice
sudo useradd -m bob
sudo passwd alice
sudo passwd bob
```

Create Group

```
sudo groupadd students
```

Add Users to Group

```
sudo usermod -aG students alice
sudo usermod -aG students bob
```

Create Shared Directory

```
sudo mkdir /shared
```

Change Ownership

```
sudo chown root:students /shared
```

Apply Permissions

```
sudo chmod 2775 /shared
```

Verify

```
ls -ld /shared
```

Students can now collaborate within the shared directory.

Permission Troubleshooting

Error

Permission denied

Check:

```
ls -l
```

Verify:

- Ownership
- Group Membership
- Permissions
- ACL Rules

Important Commands Summary

```
whoami
id
groups
useradd
adduser
passwd
userdel
groupadd
groupdel
usermod
gpasswd
chown
chgrp
chmod
getfacl
setfacl
ls -l
ls -ld
```